DATA1030 Midterm Report

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Team: Placeholder
Github Link: https://github.com/jiangisabella/data1030-placeholder

Introduction

In the United States, a union election is a formal legal process conducted by the National Labor Relations Board (NLRB) to form or disband a union in the workplace. Through certification elections, workers can legally recognize their collective bargaining power, thereby help build a more equitable workplace.

As future workers who may unionize, our team wants to understand the complex factors behind this seemingly straightforward process and identify how election duration or parent union affiliation may influence the election result. For labor organizers, examining correlation in past data also yields insights for strategizing.

Prior work on NLRB certification elections, mostly regression-based, links outcomes to unit size, petition-to-election delay, election type, region, union characteristics, and employer opposition such as consultants and unfair labor practice charges (Cooke 1983; Farber 1999; Freeman & Kleiner 1990). Unit size is the most consistent finding: from 1952 to 1998, unions won through small units more often than in large ones, and the gap widened over time (Farber 1999). Malin and Werner find that the April 2015 rule changes sped up processing but had no significant effect on outcomes (9). Contrary to common assumption, petition-to-election delay also had no significant effect (17), while unit size was a stronger predictor (7). Howard et al. report that union win rates rose from 2008 to 2024 despite swings across presidential administrations, consistent with our data.

Building on these in-sample regression analyses, we will explore an out-of-sample predictive model for union election results. This predictive model may help union organizers identify their strengths and weaknesses in current campaigns, ultimately leading to more union wins.

Data

Our data comes from the National Labor Relations Board (NLRB). Data was scraped in a two-step process: high-level case information is scraped through NLRB's election report page (“Election Reports”), while case-specific information is scraped through each case's docket activity portal (“Case Search”).

In the first step, each monthly election report for fiscal years 2012–2026 was downloaded with requests and BeautifulSoup. A fiscal year includes dates from October 1st to September 30th of the current year. For example, FY2013 runs from October 1, 2012 to September 30, 2013. Each report’s tables were extracted with pdfplumber. Each row represents one closed election, with the case number, employer, location, union, number of eligible voters, vote counts and outcome. Nine image-only reports (April 2019

and February-September 2023) could not be parsed. October 2025 has no report because of the government shutdown.

For each of the 18,687 case numbers, we also scraped its docket activity portal for every document filed, filing dates, and the list of participants. We joined case search data with election reports data using case numbers as unique identifiers. This forms our raw dataset.

To clean our data, we:

- Kept only RC (Certification of Representative Petition) cases, where workers petition to form a union; RD cases remove a union, and the employer files RM cases. The latter election types are unrelated to our machine learning problem.
- Excluded 654 multi-union elections (4.1%), since the outcome is a choice between unions and the reported vote columns only relate to the first union.
- Kept each election as a row. One case can hold multiple votes for different units, or rerun elections. We dropped rows in multi-unit cases and those marked both won and lost, only where the outcome contradicted the vote.
- Created a list of 34 parent unions from 7,983 unions, and assigned them using keywords.
- Mapped the states to regions based on the US census bureau.

The cleaned dataset has 15,270 elections from 15,241 cases and 21 columns (excluding the target variable column) (Appendix A), including:

- Continuous (`eligible`, `days_petition_to_election`, `n_employer_lawyers`, `n_petitioner_lawyers`, `n_participants`)
- Ordinal (`fiscal_year`)
- Categorical (`state`, `census_region` (5 U.S. regions), `parent_union` (34 groups))
- Binary Categorical (`employer_has_lawyer`, `petitioner_has_lawyer`)

Our target variable is `union_won`. It is defined per election and indicates how the NLRB closed the case. In our data, 70.8% of elections were won:

- 1 if the case closed with a Certification of Representative (win). This occurs with a majority of valid votes cast.
- 0 for a Certification of Results (loss), which includes tied votes.

The key features are `days_petition_to_election`, the number of days from the petition to the vote (median 31); `eligible`, the number of eligible voters (median 22); and the flag `employer_has_lawyer` (true for 88% of elections).

Exploratory Data Analysis

To clean the data, we standardized headers across years and removed repeated months from overlapping reports. The target variable `union_won` comes from the report's closed reason; if it was missing, the

WON/LOSS column was used instead. The docket data also contained the certification the NLRB issued, which served as a check. We dropped 61 rows where the docket disagreed with the outcome, or if it was a multi-unit case with both certificate types and whose label contradicted the tally. We also dropped columns that were only available post-vote, such as the closed reason, closing date, and vote counts.

If the number of eligible voters was 0 or less than the number of votes, it was treated as missing data. We calculated the days from petition to election. If the petition date was missing, we used the file date, which was only available in earlier reports. If the election date was missing, we filled it in with the first ballot tally, which usually falls on election day.

Exploratory data analysis began by checking basic dataset information, including data types and null counts, to ensure that the data was usable for modeling. We then visualized each feature on its own and against the target variable, `union_won`, to see which features would best fit for a predictive model. Below are a few features that stood out.

Unions (the petitioners) won noticeably less often when the employer had a lawyer, while the petitioner's lawyer made little difference. Legal representation could be a major contributor to a union's win or loss, so it deserves a closer look, especially as it also interacts with other features such as time.

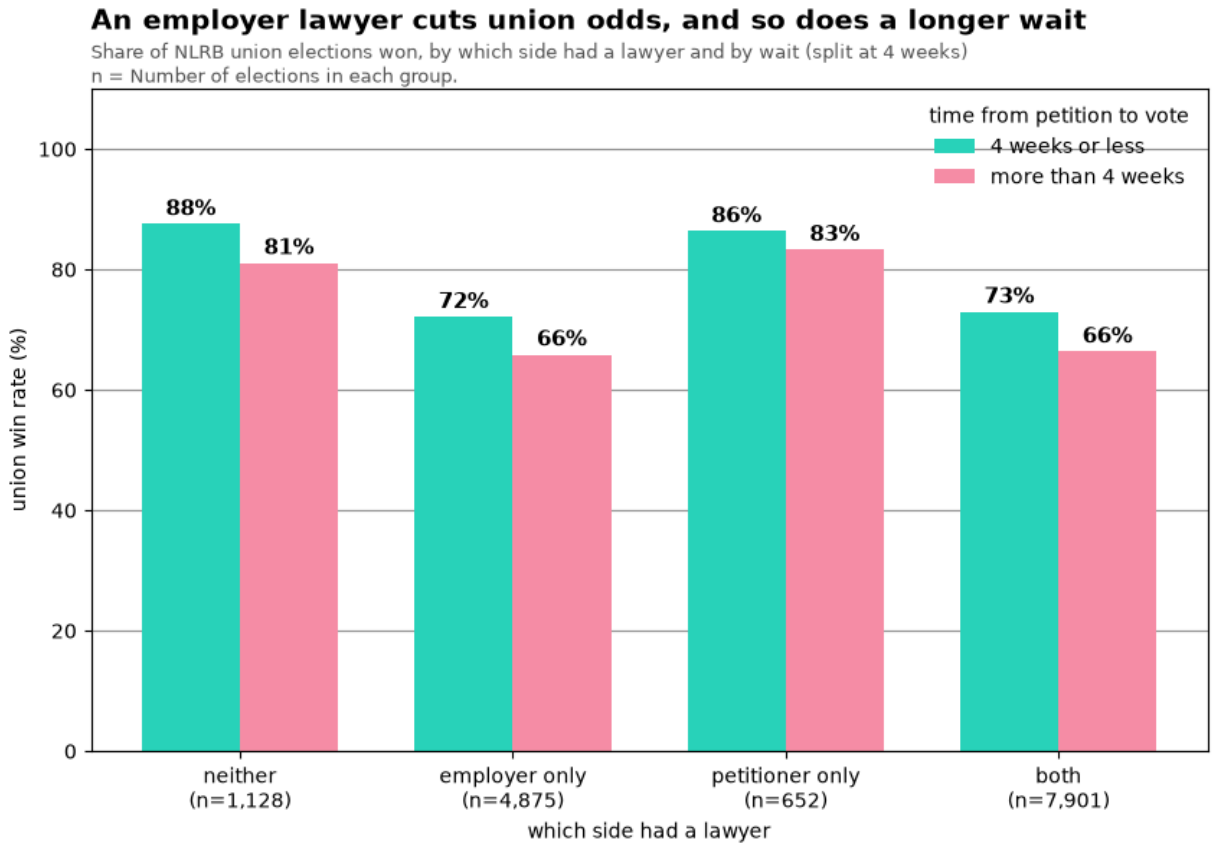


Figure 1 : Union Win Rate by Legal Representation.

It's commonly assumed that the faster a case is handled, the more likely the union is to succeed (Malin and Werner). We then visualized the time between the petition and election date (Figure 2). Elections were grouped into bins from two weeks or less to more than twelve weeks. Unions won 86% of elections held within two weeks of the petition, and win rates declined as more time passed.

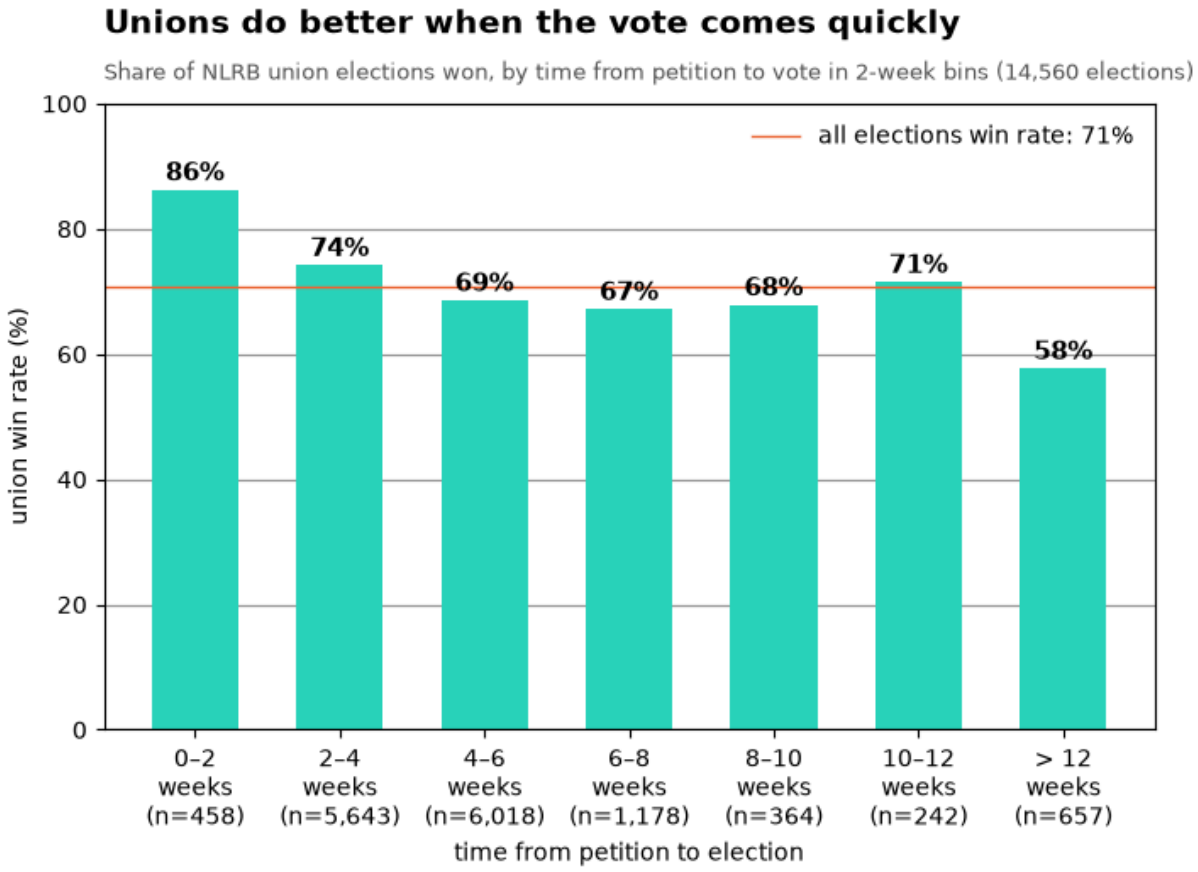


Figure 2: Union Win Rate by Time from Petition to Election

Although this chart adds no new predictive information, it is important to see the changes over the years in how many cases were won. The data shows that the union win rates increased from 63% in 2012 to 77% in 2026. The dip in 2020 likely relates to the pandemic, when a significantly lower number of elections were held.

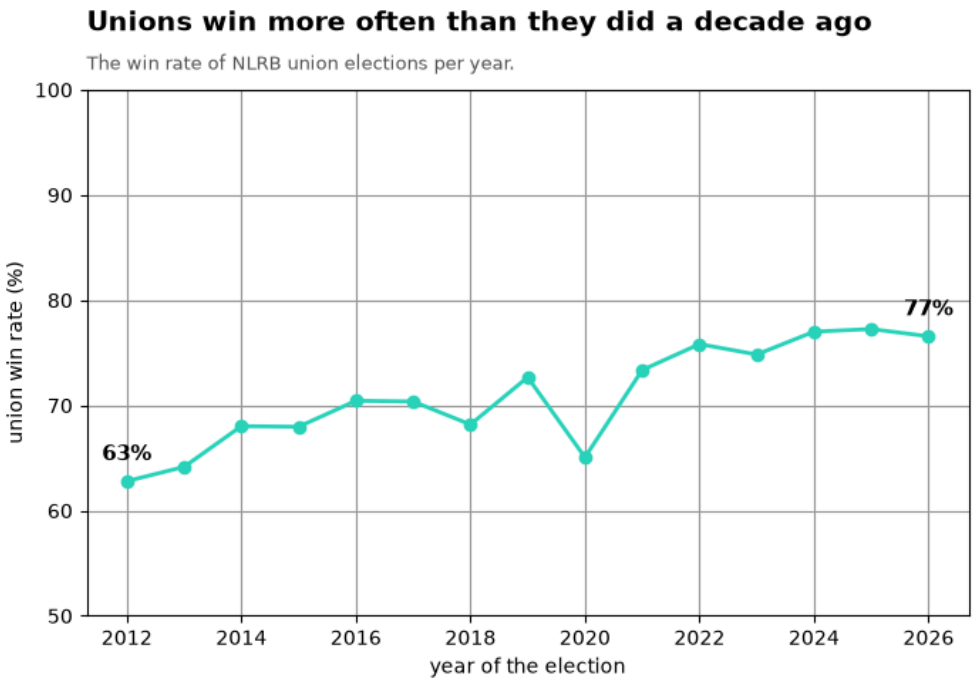


Figure 3: Union Win Rate by Year

Finally, we examined the effect of unit size on union outcomes, where unit size is the number of eligible voters. The median size of the unit was 22, while the average is 63 voters. A smaller voting pool indicated a higher chance of the union winning (Figure 4). A smaller unit may bargain more effectively, though this is just speculation.

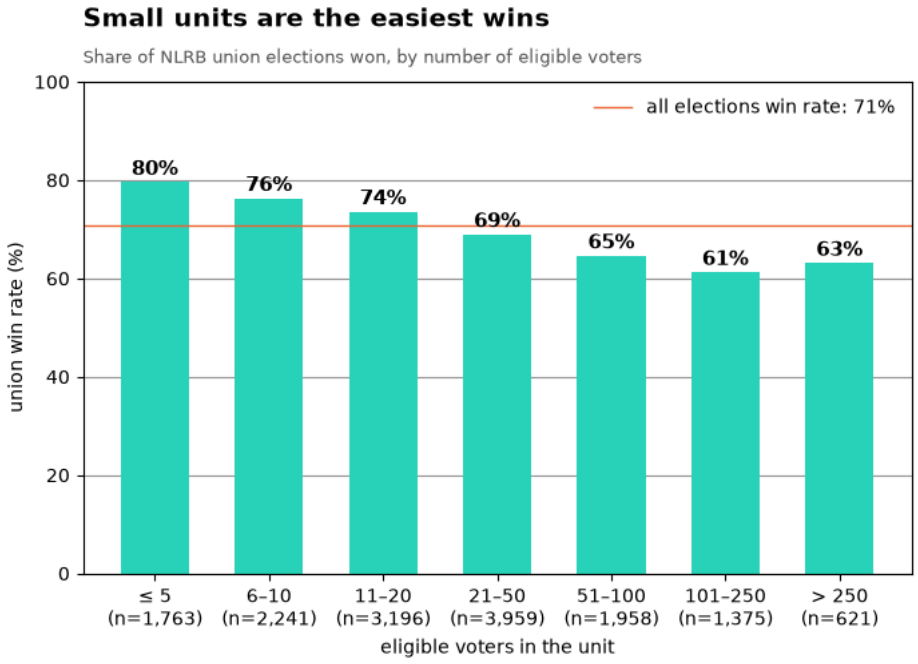


Figure 4: Union Win Rate by Unit Size

Problem Description and Potential Impact

Our goal is to predict the final legal outcome for new single-union RC elections in the United States. Our question is: **Will the single-union RC election receive a Certification of Representative or not?** This is a supervised binary classification problem, where our target variable union_won is a binary flag where 1 represents a Certification of Representative (union won) and 0 means a Certification of Results (union lost).

Our population is single-union RC elections from the NLRB fiscal years 2012-2026 (running October 2011 to September 2026). The model would be used once an election is scheduled. To avoid leakage, the feature columns are information that is known before the vote. The rows are not independent, as the union outcome likely depends on the employer. For example, employers share lawyers, or a strategy against unions. When splitting data, we will group by employer. Win rates have increased over time, it could be due to factors such as administration and rule changes. Thus, the test set will include the most recent data.

Eventually, our machine learning model can be a tool for labor organizers to strengthen their union campaign. Organizers can plug in features about their own campaign - unit size, employer practices, parent union, or geographic location - to estimate their campaign's outcome probability against 14 years of data points. This predicted process is not meant to be deterministic, but to help organizers improve their campaign, altogether increasing the odds of winning an election.

References

Data Sources

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Richard B. Freeman and Morris M. Kleiner, "Employer Behavior in the Face of Union Organizing Drives," NBER Working Paper 2805 (1988), https://doi.org/10.3386/w2805.

Appendix

Appendix A: Data Dictionary

elections_clean.csv: one row per single-union RC election, 15,270 rows x 22 columns.

fiscal_year: (ordinal) NLRB fiscal year of the monthly report the election appears in.
A fiscal year runs from October to September (FY2012 = Oct 2011 - Sep 2012).
Ordered from earliest to latest.

2012
2013
...
2026

case_number: (categorical) a unique string assigned by NLRB to identify each election case.
format: region-case type-number (*)

01-RC-216927
...

case_type: (categorical) type of NLRB representation case (*)

RC workers petition to certify a union

case_name: (categorical) a unique descriptor assigned by NLRB to identify each election case (*)

employer: (categorical) the employer of which the union election takes place (*)

city: (categorical) the city in which the union election takes place (*)

state: (categorical) the state in which the union election takes place

AL alabama
AK alaska
...
WI wisconsin
WY wyoming

census_region: (categorical) groups states into four geographic regions according to U.S. Census Bureau conventions

- Northeast
- Midwest
- South
- West
- Territory

union_name: (categorical) name of union that the election is voted under (*)

parent_union: (categorical) parent union that the election union is part of

Teamsters	International Brotherhood of Teamsters
SEIU	Service Employees International Union, incl. 1199 locals
IAM	International Association of Machinists and Aerospace Workers
IUOE	International Union of Operating Engineers
UFCW	United Food and Commercial Workers
IBEW	International Brotherhood of Electrical Workers
Workers United	Workers United (SEIU affiliate, kept as its own group)
USW	United Steelworkers
CWA	Communications Workers of America, incl. NewsGuild
SPFPA	Security, Police and Fire Professionals of America
UAW	United Auto Workers
LIUNA	Laborers' International Union of North America
AFSCME	American Federation of State, County and Municipal Employees
Nurses	any union with "nurse" in its name
IATSE	International Alliance of Theatrical Stage Employees
UA (Plumbers)	United Association of plumbers, pipefitters and sprinkler fitters
ATU	Amalgamated Transit Union
UNITE HERE	UNITE HERE (hotel and food service workers)
RWDSU	Retail, Wholesale and Department Store Union (UFCW affiliate, kept as its own group)
SMART	Sheet Metal, Air, Rail and Transportation Workers
AFT	American Federation of Teachers
OPEIU	Office and Professional Employees International Union
UGSOA	United Government Security Officers of America
IUPAT	International Union of Painters and Allied Trades
BCTGM	Bakery, Confectionery, Tobacco Workers and Grain Millers
Carpenters	United Brotherhood of Carpenters
UWUA	Utility Workers Union of America
IAFF	International Association of Fire Fighters

ILWU	International Longshore and Warehouse Union
NATCA	National Air Traffic Controllers Association
Ironworkers	International Association of Ironworkers
TWU	Transport Workers Union
Other	smaller and independent unions

eligible: (continuous) count of eligible voters in a union election

petition_date (yyyy-mm-dd): earliest date on which the union election is petitioned to the NLRB. file_date if the docket lists none. (*)

election_date (yyyy-mm-dd): date on which the union election takes place.
Where the report left it blank, it is filled from the docket's first Tally of Ballots (*)

days_petition_to_election: (continuous) duration between election date and petition date, counted in days

n_participants: (continuous) number of parties and representatives who participated in the case (employer, union, lawyers, intervenors)

employer_has_lawyer: (binary categorical) whether the employer is supported by lawyers against the election

1 the employer has lawyers
0 the employer doesn't have lawyers

n_employer_lawyers: (continuous) number of lawyers the employer has against the election

employer_law_firms: (categorical) names of the employers' lawyers' law firms. Several values, separated by ;.

petitioner_has_lawyer: (binary categorical) whether the union (petitioner) is supported by lawyers for the election

1 union supported by lawyers
0 unions not supported by lawyers

n_petitioner_lawyers: (continuous) number of lawyers the union has for the election, as a count

has_intervenor: (binary categorical) describes the presence of intervenors during the election

- 1 has intervenors
- 0 does not have intervenors

union_won: (binary categorical) [Target variable] describes the final official union election result based on vote counts

- 1 win: Certification of Representative issued
- 0 loss: Certification of Results issued

(*) are descriptive variables that will likely not be used in the ML pipeline